

L0 — plan.md Results Summary (Level-0 warm-up)

The four **plan.md** legs after Data Prep, run on the 450-clip inventory. **One question:** is YAMNet's embedding space good enough for our classes, or does it need backbone work? **Answer: the embedding space is excellent; only the head needs adapting.**

Where everything lives

What	Path
Plots (PNG)	<code>experiments/outputs/figures/</code>
Per-clip top-5, metrics, embeddings	<code>experiments/outputs/*.csv, *.json, yamnet_features.npz</code>
Scripts	<code>experiments/scripts/{extract_yamnet, leg1_baseline, leg2_embeddings, leg3_linear_probe, leg4_mlp_head}.py</code>
Per-leg reports	<code>docs/L1-L4</code>

Findings at a glance

Leg	Result	Plot
L1 baseline	Raw 521-class head useless (Elephant→Jackhammer, Frog→Eruption; 0–12% relevant in top-5)	<code>baseline_top1_hist.png</code>
L2 embeddings	kNN purity 0.80; Drone/Lion/Frog/Monkey cluster cleanly; Elephant overlaps background	<code>embeddings_2d.png</code>
L3 linear probe	AUC 0.90 (Elephant) → 1.00 (Drone/Lion/Monkey), Frog 0.99 — all PASS	<code>roc_linear_probe.png</code>
L4 MLP head	No improvement over linear ($\Delta \leq 0$); Elephant overfits	<code>mlp_loss_curves.png, roc_mlp_head.png</code>

The throughline

1. The **head** knows nothing about our classes out of the box (L1).
2. But the **embeddings** separate them almost perfectly (L2, L3) — the head is the only problem.
3. So a **linear probe is at ceiling**, and added non-linearity only overfits (L4).
4. ⇒ Effort belongs in **features/data realism**, not model capacity. **Elephant** is the one class that's genuinely entangled with background and needs targeted help.

Caveat → hand-off to the bigger plan (`experiments.pdf`)

These are **clean library clips**, so this is an *upper bound*, not a deployment estimate. The real program trains/evaluates on **post-ODAS** reconstructed spectra (beamformed, Griffin-Lim artefacts, ambient ghost-track false positives). What L1–L4 establish for that program:

- YAMNet embeddings are the right representation — proceed with the head/fine-tune approach.
- Watch **Elephant** (low-freq rumble ≈ ambient) and **false positives from background** — exactly the focus of `experiments.pdf` Phase 1 (ambient ablation) and Phase 2 EXP-B4 (hard negatives).

See `docs/README.md` for the full step/leg index.